

PORT ISABEL CAMERON COUNTY AP, TX, USA

WMO: 722508

Lat: 26.166N

Lon: 97.346W

Elev: 6

StdP: 101.26

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 12957

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.9	6.0	-3.4	2.8	9.6	-0.6	3.6	10.3	12.8	20.5	11.9	16.2	7.0	330	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	35.2	26.1	34.4	26.1	33.7	26.1	27.5	32.0	27.2	31.6	26.9	31.2	7.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	28.9	26.2	21.6	28.9	25.8	21.1	29.0	87.1	32.0	85.6	31.6	84.3	31.4	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.9	9.8	DB	1.1	37.5	2.0	1.6	-0.3	38.6	-1.5	39.6	-2.6	40.5	-4.0	41.6	
			WB	-0.7	28.2	1.7	0.7	-1.9	28.7	-2.9	29.0	-3.9	29.4	-5.2	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	16.1	18.2	21.0	24.2	27.1	29.2	29.6	30.0	28.3	25.3	20.9	17.6
	DBStd	5.93	4.85	5.07	4.06	3.13	2.31	1.47	1.24	1.32	1.83	3.35	4.47	4.79
	HDD10.0	21	10	5	1	0	0	0	0	0	0	0	1	4
	HDD18.3	291	99	61	23	4	0	0	0	0	0	2	27	73
	CDD10.0	5124	199	234	343	425	529	574	609	619	548	474	329	239
	CDD18.3	2352	31	57	107	179	271	324	351	361	298	218	105	51
	CDH23.3	24685	117	234	483	1256	2742	4040	4511	4726	3478	2169	694	234
(14) Wind	WSAvg	5.1	4.9	5.4	5.8	6.1	6.2	5.5	5.3	4.8	4.1	4.4	4.6	4.8
	PrecAvg	635	26	28	34	40	50	63	47	50	136	92	45	35
	PrecMax	833	93	106	187	216	154	196	234	156	320	258	159	124
	PrecMin	363	0	1	3	0	0	10	3	0	27	0	0	3
	PrecStd	165	23	26	41	50	38	47	55	39	79	73	38	30
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.7	29.8	30.6	33.0	34.0	35.7	35.7	36.3	35.3	33.2	30.8	28.8
		MCWB	20.7	21.0	23.2	23.8	25.4	25.9	26.0	26.4	26.2	25.5	24.2	22.7
	2%	DB	26.1	27.8	28.6	31.2	32.8	34.2	34.7	35.3	34.1	32.2	29.3	27.3
		MCWB	20.3	21.8	22.2	23.4	25.2	26.2	26.1	26.3	26.1	25.3	23.3	22.3
	5%	DB	24.6	26.2	27.4	29.8	31.9	33.4	33.9	34.5	33.0	31.3	28.1	26.0
		MCWB	20.0	21.1	21.7	23.4	25.1	26.0	26.1	26.2	25.9	25.0	22.8	21.7
	10%	DB	23.0	24.5	26.1	28.5	30.8	32.6	33.2	33.7	32.1	30.3	27.1	24.4
		MCWB	19.2	20.4	21.3	22.9	24.7	25.9	26.1	26.2	25.8	24.4	22.5	21.1
	0.4%	WB	22.7	23.3	24.0	25.9	26.9	28.0	27.8	28.0	27.7	27.1	25.7	24.2
		MCDB	25.0	27.1	27.8	29.6	30.8	33.1	32.1	32.3	32.3	30.3	28.3	26.3
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.9	22.6	23.3	25.2	26.4	27.4	27.3	27.5	27.1	26.6	24.9	23.3
		MCDB	24.3	25.9	26.6	29.0	30.3	31.8	31.6	32.2	31.6	29.7	27.6	25.8
	5%	WB	21.1	21.9	22.8	24.5	25.9	26.9	26.9	27.1	26.8	26.0	24.2	22.6
		MCDB	23.5	24.6	25.8	28.0	29.6	31.1	31.2	31.8	31.1	29.3	26.8	25.0
	10%	WB	20.3	21.4	22.3	24.0	25.5	26.6	26.7	26.8	26.4	25.6	23.4	21.8
		MCDB	22.6	23.8	25.0	27.2	29.2	30.7	31.0	31.5	30.6	28.9	26.0	23.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	8.6	8.1	7.7	7.1	7.2	7.4	7.9	7.3	8.3	8.8	9.0
		MCDBR	10.2	9.3	8.2	8.6	7.7	7.7	8.1	8.5	8.1	8.1	8.6	9.1
		MCWBR	5.8	4.4	3.7	3.5	2.4	2.2	2.1	2.1	2.7	3.3	4.2	5.1
	5% WB	MCDBR	8.7	7.8	7.3	7.2	6.7	6.9	7.3	7.9	7.2	7.1	7.2	7.9
		MCWBR	5.8	4.3	3.7	3.4	2.4	2.5	2.4	2.5	2.7	3.5	4.1	5.2
(40) Clear-Sky Solar Irradiance	taub		0.369	0.382	0.401	0.448	0.464	0.434	0.454	0.439	0.443	0.396	0.380	0.372
	taud		2.465	2.421	2.364	2.231	2.220	2.371	2.324	2.375	2.325	2.459	2.463	2.457
	Ebn at Noon		880	895	894	856	838	857	840	854	842	870	862	859
	Edh at Noon		98	110	122	143	145	124	129	122	126	104	98	95
(44) All-Sky Solar Radiation	RadAvg		3.08	3.72	4.61	5.50	6.13	6.61	6.57	6.28	5.22	4.58	3.50	2.79
	RadStd		0.34	0.54	0.47	0.46	0.37	0.24	0.44	0.26	0.36	0.41	0.36	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105

Nomenclature: See separate page